

Internship Task Submission (traffic data analysis)

Name: S mohamed sohail

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Traffic Data Analysis

1. Introduction

Traffic data analysis is essential for understanding traffic flow, congestion patterns, and peak traffic hours. This analysis helps in improving traffic management, road safety, and urban planning.

2. Problem Statement

The objective of this project is to analyze traffic data to identify traffic patterns, peak congestion periods, and trends that can assist in better traffic control and planning.

3. Dataset Description

The dataset consists of traffic-related information such as date and time, vehicle count, traffic density, and other relevant attributes used to study traffic behavior.

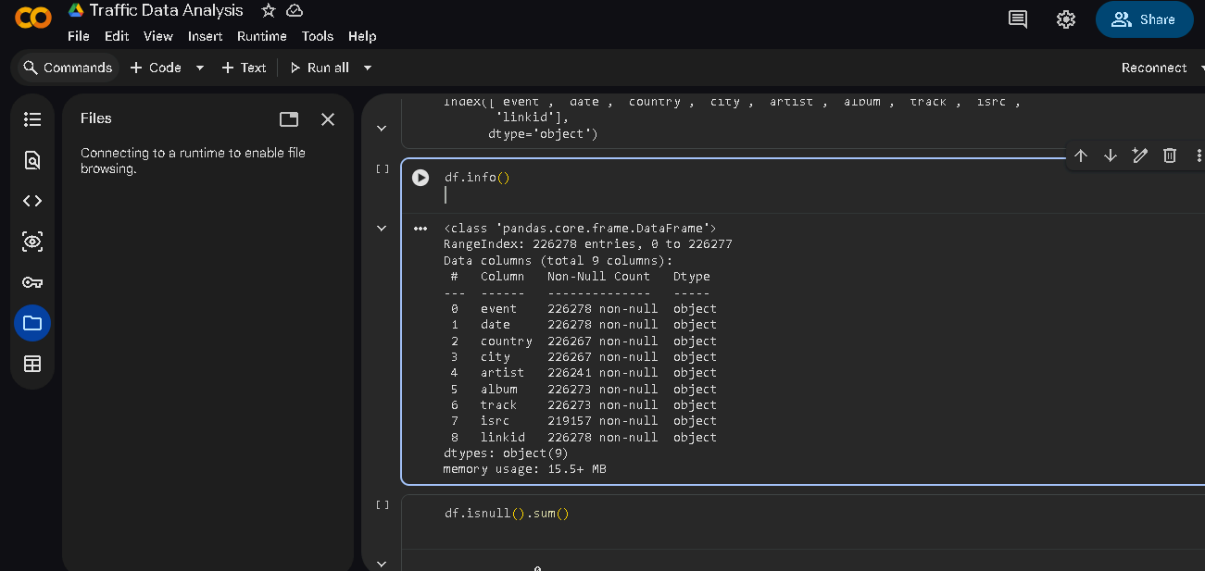
4. Tools & Technologies Used

- **Python**

- Google Colab
 - Pandas
 - Matplotlib
 - Google Drive
 - GitHub
-

5. Data Preprocessing

The dataset was cleaned by handling missing values, removing duplicates, and converting date-time columns into appropriate formats for analysis.



```
index([ event , date , country , city , artist , album , track , isrc ,
      'linkid'],
      dtype='object')

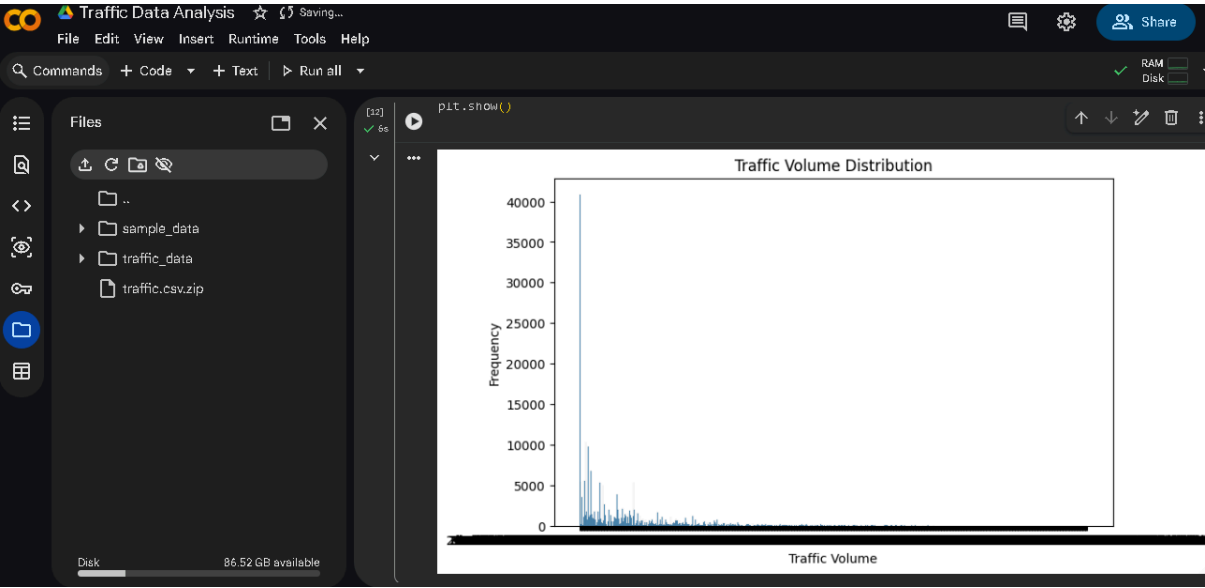
[] df.info()

... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 226278 entries, 0 to 226277
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0    event      226278 non-null  object
1    date        226278 non-null  object
2    country     226267 non-null  object
3    city        226267 non-null  object
4    artist      226241 non-null  object
5    album       226273 non-null  object
6    track       226273 non-null  object
7    isrc        219157 non-null object
8    linkid      226278 non-null  object
dtypes: object(9)
memory usage: 15.5+ MB

[] df.isnull().sum()
```

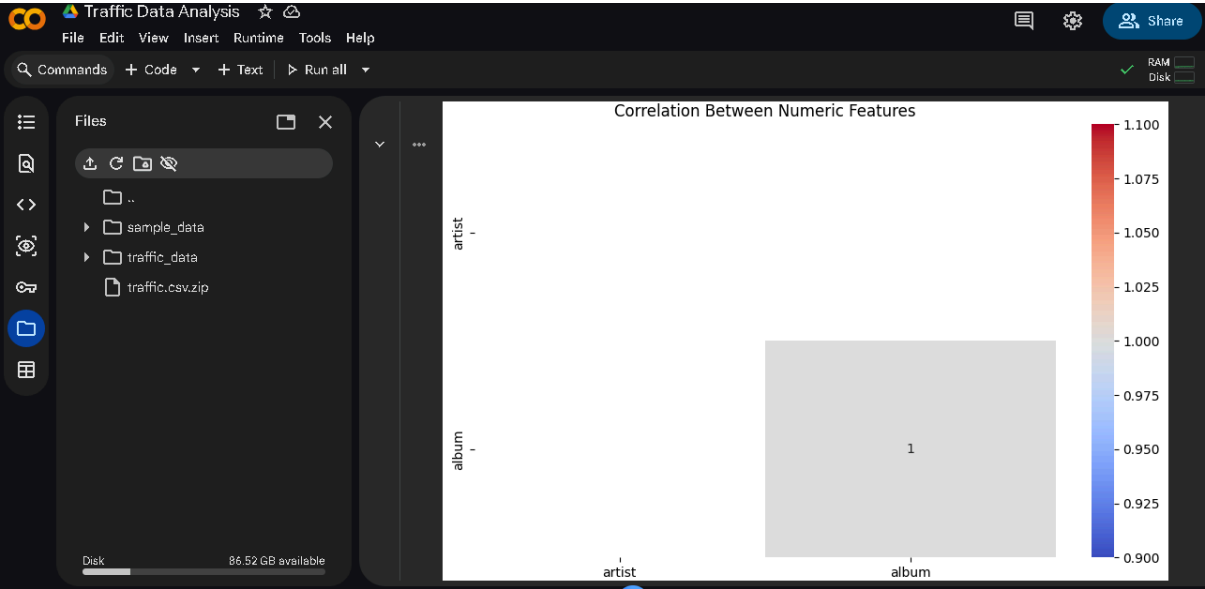
6. Exploratory Data Analysis

Traffic patterns were analyzed using visualizations to understand peak traffic hours and congestion levels.



7. Analysis & Findings

The analysis revealed specific peak hours with high traffic volume and variations in traffic flow across different time periods.



8. Practical Applications

- Traffic signal optimization
 - Congestion control during peak hours
 - Urban infrastructure planning
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9. Conclusion

This project successfully analyzed traffic data to identify congestion trends and peak traffic periods, helping improve traffic management strategies.

10. GitHub Repository Link

Paste your GitHub link here:

<https://github.com/sohailgithub442/traffic-data-analysis/tree/main>